CEN/TC 434 WG1 · EN 16931-1:2026 AMENDMENT

Gross prices in EN 16931-1:2026

Calculate from the agreed price, round only once — issues #164 and #163.

Slides as PDF

Svante Schubert
Editor of the syntax bindings · CEN/TC 434 WG3 · for DIN

COMMON GROUND

What we all want

In the end, we all want accurate data — and these goals do not exclude each other.

Correct VAT, to the cent

- Every invoice reproduces the agreed price
- VAT exact to the cent, as tax law requires
- “Accounting is 100% correct or it is not” (Paul Simons)

Working software keeps working

- Invoices with 2 decimal places stay conformant
- “Not to force companies” (Cyrille Sautereau)
- 5 million companies in France, 1.2 million in Belgium

A STABLE STANDARD

- The core is what every receiver accepts
- No requirement on software: no new minimum of decimal places
- Additions are optional paths; the net model stays as it is
- Syntax unchanged: amounts are xsd:decimal in UBL and CII

Goal of this session: two decisions for WG1 — gross prices in the core (#164) and no forced rounding (#163)

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LEGAL BASIS

What the law requires

The invoice must reproduce the consideration and the VAT exactly.

VAT to the cent

- Calculated to the cent, rounded commercially (BMF letter of 15 Oct 2026, para. 17; UStAE 14.5 (20))
- Net amount = consideration minus VAT (§ 10 (1) UStG)

Proportional and exact

- VAT “exactly proportional to the price” (CJEU Ahold, C-484/06, para. 38)
- Rounding only where necessary, “as closely as possible” (para. 39)

INVOICE CONTENT AND REPORTING

- Art. 226(8): taxable amount per rate, unit price exclusive of VAT
- Art. 206, 168: the seller pays the VAT, the buyer deducts it
- New Art. 232 (ViDA): an EN 16931 invoice needs no acceptance
- New Art. 263(3) (ViDA): DRR accepts data that complies with the EN

Sources: VAT Directive 2006/112/EC · ViDA, Directive (EU) 2025/516 · CJEU C-484/06 · BMF letter of 15 Oct 2026

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EN 16931-1:2026 · TODAY

The specification contradicts itself

6.5.14 forbids intermediate rounding — Table 28 forces it.

6.5.14: round final results only

- "… on the final calculation results and not on any intermediate results"
- The same principle was already in the 2017 edition

Table 28: maximum decimal places

- BT-146 Item net price: Minor Unit + 2 (new in 2026)
- BT-131 Invoice line net amount and all sums: Minor Unit
- BT-129 Invoiced quantity: 4 dp, BT-149: 2 dp (new in 2026)

CONSEQUENCES

- Every rule starts from rounded values — the errors add up
- Tolerances of up to €1 (BR-S-9) hide the deviation
- BMF: the standard and the tax rounding rule cannot both be met (para. 18)
- Next: the calculation chains of 2017 and 2026 and their worst case

2017: only amounts were limited — prices and quantities had no maximum (Table 26)

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CALCULATION CHAIN · EN 16931-1:2017

From gross price via net price to VAT

One intermediate rounding: **BT-131 to 2 decimal places**

INPUT & INTERMEDIATE VALUES

BT-146	**Item net price** = BT-148 − BT-147	unlimited unrounded	± 0 stated exactly
BT-131	**Invoice line net amount** = BT-146 × BT-129 ÷ BT-149 + Σ BT-141 − Σ BT-136	2 dp interim rounding	± €0.005 per line
BT-106	**Sum of invoice line net amount** = Σ BT-131	2 dp Σ of rounded values	± €0.50 × 100 lines

FINAL RESULTS

BT-116	**VAT category taxable amount** = Σ BT-131 + Σ BT-99 − Σ BT-92	2 dp interim rounding	± €0.50 Σ of rounded lines
BT-117	**VAT category tax amount** = BT-116 × BT-119 ÷ 100	2 dp interim rounding	± €0.10 €0.50 × 19%
BT-110	**Invoice total VAT amount** = Σ BT-117	2 dp Σ of rounded values	± €0.10
BT-115	**Amount due for payment** = BT-110 + BT-109 Invoice total amount without VAT − BT-113 Paid amount + BT-114 Rounding amount	2 dp final value · bank	± €0.60 €0.50 + €0.10

ALLOWED DECIMAL PLACES

BT-148 Item gross price	unlimited
BT-147 Item price discount	unlimited
BT-146 Item net price	unlimited
BT-129 Invoiced quantity	unlimited
BT-149 Item price base quantity	unlimited
BT-141 Invoice line charge or tax amount	2 dp
BT-136 Invoice line allowance amount	2 dp
BT-131 Invoice line net amount	2 dp
BT-106 Sum of invoice line net amount	2 dp
BT-99 Document level charge or tax amount	2 dp
BT-92 Document level allowance amount	2 dp
BT-119 VAT category rate (law)	unlimited

DECIMAL PLACES · FINAL VALUES

BT-116 VAT category taxable amount	2 dp
BT-117 VAT category tax amount	2 dp
BT-110 Invoice total VAT amount	2 dp
BT-115 Amount due for payment	2 dp

Worst case with number of lines and quantity:
→ "How large does the deviation get?"

Red = rounded intermediate value Green = paid amount, rounded once (bank) Orange = rounded input value
dp = decimal places · 2017 (Tab. 26): all amounts fixed 2 dp, regardless of the currency · ± for 100 lines of 100 units, 19% VAT
Rainbow = order of the calculation steps; a BT keeps its colour where it is reused further down

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CALCULATION CHAIN · EN 16931-1:2026

From gross price via net price to VAT

Two intermediate roundings: **BT-146 to Minor Unit + 2** **BT-131 to the Minor Unit**

INPUT & INTERMEDIATE VALUES

BT-146	**Item net price** = BT-148 − BT-147	Minor Unit + 2 interim rounding	± €0.00005 price rounded once
BT-131	**Invoice line net amount** = **BT-146** × BT-129 ÷ BT-149 + Σ BT-141 − Σ BT-136	Minor Unit interim rounding	± €0.01 €0.005 + 100 × €0.00005
BT-106	**Sum of invoice line net amount** = Σ **BT-131**	Minor Unit Σ of rounded values	± €1.00 × 100 lines

FINAL RESULTS

BT-116	**VAT category taxable amount** = Σ **BT-131** + Σ BT-99 − Σ BT-92	Minor Unit interim rounding	± €1.00 Σ of rounded lines
BT-117	**VAT category tax amount** = **BT-116** × BT-119 ÷ 100	Minor Unit interim rounding	± €0.19 €1.00 × 19%
BT-110	**Invoice total VAT amount** = Σ **BT-117**	Minor Unit Σ of rounded values	± €0.19
BT-115	**Amount due for payment** = **BT-110** + **BT-109** Invoice total amount without VAT − **BT-113** Paid amount + **BT-114** Rounding amount	Minor Unit final value · bank	± €1.19 €1.00 + €0.19

ALLOWED DECIMAL PLACES

BT-148 Item gross price	Minor Unit + 2
BT-147 Item price discount	Minor Unit + 2
BT-146 Item net price	Minor Unit + 2
BT-129 Invoiced quantity	4 dp
BT-149 Item price base quantity	2 dp
BT-141 Invoice line charge or tax amount	Minor Unit
BT-136 Invoice line allowance amount	Minor Unit
BT-131 Invoice line net amount	Minor Unit
BT-106 Sum of invoice line net amount	Minor Unit
BT-99 Document level charge or tax amount	Minor Unit
BT-92 Document level allowance amount	Minor Unit
BT-119 VAT category rate (law)	2 dp

DECIMAL PLACES · FINAL VALUES

BT-116 VAT category taxable amount	Minor Unit
BT-117 VAT category tax amount	Minor Unit
BT-110 Invoice total VAT amount	Minor Unit
BT-115 Amount due for payment	Minor Unit

Worst case with number of lines and quantity:
→ "How large does the deviation get?"

Red = rounded intermediate value **Green** = paid amount, rounded once (bank) **Orange** = rounded input value
dp = decimal places · Minor Unit = smallest currency unit, ISO 4217 (EUR: 2 dp) · ± for 100 lines of 100 units, 19% VAT
Rainbow = order of the calculation steps; a BT keeps its colour where it is reused further down

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WORST CASE · EN 16931-1:2017

How large does the deviation get?

One factor: the number of lines n — each line rounds **BT-131** Invoice line net amount and loses up to ± €0.005.

Factor 1 · price rounding × quantity Q

BT-146 Item net price is unlimited in 2017 — the price is not rounded, so the quantity magnifies no error.

`none`

Factor 2 · number of lines n

Each line rounds **BT-131** Invoice line net amount to 2 decimal places: up to ± €0.005. **BT-106** Sum of invoice line net amount and **BT-116** VAT category taxable amount add up all n lines — and with them the errors of all lines.

`± n × €0.005`

NET DEVIATION **BT-106** Sum of invoice line net amount / **BT-116** VAT category taxable amount

Units per line (Q) → ↓ Lines (n)	100 units	10,000 units	1,000,000 units
1 line	Price × quantity ± 0 + rounding BT-131 ± €0.005 = ± €0.005	Price × quantity ± 0 + rounding BT-131 ± €0.005 = ± €0.005	Price × quantity ± 0 + rounding BT-131 ± €0.005 = ± €0.005
100 lines	± €0.50	± €0.50	± €0.50
10,000 lines	± €50.00	± €50.00	± €50.00

Assumption: all roundings go in the same direction, BT-149 Item price base quantity = 1, 19% VAT.

TOTAL · NET

n × €0.005

VAT: 19% of that. Open-ended — the more lines, the larger the deviation.

100 lines of 10,000 units
Net ± €0.50 VAT ± €0.10

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WORST CASE · EN 16931-1:2026

How large does the deviation get?

Two factors: first the quantity Q × rounded **BT-146** Item net price, then the number of lines n — open-ended.

Factor 1 · price rounding × quantity Q

BT-146 Item net price is "squeezed" once into Minor Unit + 2 (EUR: 4 dp): up to ± €0.00005. **BT-131** Invoice line net amount multiplies this one error by **BT-129** Invoiced quantity. #148: €0.123449 → €0.1234; with 10,000 units €0.49 is missing.

`± Q × €0.00005 per line`

Factor 2 · number of lines n

Each line rounds **BT-131** Invoice line net amount additionally to the Minor Unit: up to ± €0.005. **BT-106** Sum of invoice line net amount and **BT-116** VAT category taxable amount add up all n lines — and with them the errors of all lines.

`± n × €0.005`

NET DEVIATION **BT-106** Sum of invoice line net amount / **BT-116** VAT category taxable amount

Units per line (Q) → ↓ Lines (n)	100 units	10,000 units	1,000,000 units
1 line	Price × quantity ± €0.005 + rounding BT-131 ± €0.005 = ± €0.01	Price × quantity ± €0.50 + rounding BT-131 ± €0.005 = ± €0.505	Price × quantity ± €50.00 + rounding BT-131 ± €0.005 = ± €50.005
100 lines	± €1.00	± €50.50	± €5,000.50
10,000 lines	± €100.00	± €5,050.00	± €500,050.00

Assumption: all roundings go in the same direction, BT-149 Item price base quantity = 1, 19% VAT. Likewise: BT-129 Invoiced quantity only 4 dp — up to ± 0.00005 × price per line for expensive items.

TOTAL · NET

`n × (Q × €0.00005 + €0.005)`

VAT: 19% of that. Open-ended — more lines or larger quantities, more deviation.

100 lines of 10,000 units
Net **± €50.50** VAT **± €9.60**

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GROSS PRICES · EN 16931-1:2026

Gross prices: often impossible today

With net amounts and VAT in cents, many agreed gross amounts cannot be reached.

Gross amount, 19% VAT	Net in cents	Net + VAT in cents
€9.99 (receipt)	8.39	8.39 + 1.59 = 9.98 ×
	8.40	8.40 + 1.60 = 10.00 ×
Not reachable at all	19 of 119	16% of all gross amounts
Invoices with 5 lines	simulation	52% miss the receipt total
Invoices with 35 lines	simulation	81% miss the receipt total

Each cent of net amount adds 1.19 cents of gross amount, so 19 of 119 gross amounts are skipped. At 7%: 7/107, at 20%: 20/120, at 21%: 21/121.

Only the tolerance of BR-S-9 lets an issuer state 8.39 + 1.60 — with a VAT that is not 19% of the net amount.

THE BMF CONFIRMS IT

Letter of 15 October 2026, section 2.5

- Para. 17: VAT to the cent, rounded commercially
- Para. 18: core and tax rounding rule cannot both be met
- Para. 19: no gross format in XRechnung, ZUGFeRD — workaround until 31 Dec 2027

The workaround keeps only the totals per rate in the structured part; the line items move to an attachment.

Examples: receipts · fixed book prices · retail and marketplaces · statutory gross prices

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ROOT CAUSE

Converting net ↔ gross never ends

Whichever price was agreed must lead — never convert it per line.

Conversion never ends

- 9.99 ÷ 1.19 = 8.394957983193… (periodic)
- 8.40 × 1.19 = 9.996 — not a cent amount
- Exact only with unlimited decimal places

Every maximum fails at some quantity

- Price cut at the d-th dp: error ≤ BT-129 × 0.5 × 10^{-d}
- 8 dp and 10,000,000 units: up to €0.05
- Cyrille Sautereau's sheet: 6 instead of 4 dp remove €0.05

THE RULE

- The price with fewer decimal places is the original one and leads
- No conversion per line: lines are net (net model) or gross (gross model)
- One conversion only: extract the VAT once per VAT rate (VAT first)
- Net values only where the law requires them

Net model: net price leads, VAT added · Gross model (#164): price plus VAT leads, VAT extracted

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AMENDMENT · PROPOSAL 1/2

Gross prices for the EN core (#164)

If the price is fixed including VAT, it should lead the invoice.

Step	Net model (today)	Gross model (new)
Price	BT-146 Item net price	BT-146V Item net price plus VAT
Line	BT-131 = price × quantity …	BT-131V = price plus VAT × quantity …
Per category	BT-116 = Σ BT-131 …	BT-116V = Σ BT-131V …
VAT	BT-117 = BT-116 × rate added	BT-117 = BT-116V × rate ÷ (100 + rate) extracted
Net per category	(given)	BT-116 = BT-116V − BT-117

BT-146V on each line selects the model; an indicator may confirm it. BT-146 remains mandatory (Art. 226(8) VAT Directive), in the gross model for information only. Not for categories N, D, F, I, J.

Often impossible today: At 19% VAT, 16% of all gross amounts cannot be reached with net and VAT in the Minor Unit — €9.99 = 8.39 + 1.59 → 9.98. With 5 lines 52%, with 35 lines 81% miss the receipt total.

EXAMPLE: FIXED BOOK PRICES

1,000 books at €24.95 incl. 7% VAT

	Total	VAT
Target	24,950.00	1,632.24
Net 2026	24,950.05	1,632.25
Gross model	24,950.00	1,632.24

Net model: 5 cents too much, VAT 1 cent too high. Gross model, VAT first: €1,632.24299… rounded once to €1,632.24; the net amount is the remainder.

Use cases: **fixed book prices · receipt → invoice · retail and marketplaces · statutory gross prices**

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GROSS MODEL (#164) · DATA FLOW

One data flow, two directions

Same steps, mirrored business terms — they differ only where the VAT enters (orange).

NET MODEL · BT-146 LEADS	SHARED	GROSS MODEL (#164) · BT-146V LEADS
BT-146[1] = BT-148 − BT-147	**Price**	**BT-146V** = BT-148V − BT-147V · BT-146[1] for information
BT-131 = BT-146 ÷ BT-149 × BT-129 + Σ BT-141 − Σ BT-136	**Line** · BT-129[1], BT-149	**BT-131V** = BT-146V ÷ BT-149 × BT-129 + Σ BT-141V − Σ BT-136V
BT-136[1], BT-141 on the bases BT-137, BT-142	**Line A/C** · BT-138, BT-143	BT-136V, BT-141V on the bases BT-137V, BT-142V
BT-92[1] = BT-94 × BT-93 ÷ 100 · BT-99 = BT-101 × BT-100 ÷ 100	**Document A/C** · BT-94, BT-101	**BT-92V** = BT-94 × BT-93V ÷ 100 · BT-99V = BT-101 × BT-100V ÷ 100
BT-106 = Σ BT-131 · BT-107 = Σ BT-92 · BT-108 = Σ BT-99	**Sums**	**BT-106V** = Σ BT-131V · BT-107V = Σ BT-92V · BT-108V = Σ BT-99V
BT-116 = Σ BT-131 + Σ BT-99 − Σ BT-92	**VAT category** · BT-118, BT-119[2][3]	**BT-116V** = Σ BT-131V + Σ BT-99V − Σ BT-92V
BT-117 = BT-116 × BT-119 ÷ 100 · added	**VAT** · BT-117[3]	**BT-117** = BT-116V × BT-119 ÷ (100 + BT-119) · extracted
BT-116 from the lines, see above	**Taxable amount** · BT-116[1][3]	**BT-116** = BT-116V − BT-117 · the remainder (VAT first)[4]
BT-109 = BT-106 − BT-107 + BT-108 · BT-112 = BT-109 + BT-110	**Totals** · BT-110[2] = Σ BT-117	**BT-112** = BT-106V − BT-107V + BT-108V · BT-109 = BT-112 − BT-110
Payment · BT-115 = BT-112 − BT-113 + Σ BT-179 + BT-114 — the same in both models; BT-114 shows the one rounding		

net term · "plus VAT" counterpart · shared · where the models differ · A/C = allowances and charges · [1] [2] [3] [4] required by law:

[1] VAT Directive 2006/112/EC, Art. 226(6) and (8): quantity, unit price exclusive of VAT, discounts not included in it, taxable amount per rate

[2] Art. 226(9) and (10): VAT rate and VAT amount payable (EN 16931-1:2026, Annex B)

[3] Germany, § 14 (4) no. 7 and 8 UStG: net amount per rate, rate and tax amount — the tax amount calculated to the cent and rounded commercially, half up (UStAE 14.5 (20); BMF letter of 15 Oct 2026, para. 17)

[4] § 10 (1) UStG: the net amount (Entgelt) is the consideration minus the VAT

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GROSS MODEL (#164) · EXAMPLE

Fuel: the tenth of a cent is in the price

The amounts that are paid, booked and declared stay in cents.

Business term	Value	Origin
BT-146V Item net price plus VAT	1.799	price at the pump
BT-129 Invoiced quantity	45.37 l	dispensed
BT-131V Invoice line net amount plus VAT	81.62	81.62063, paid as 81.62
BT-117 VAT category tax amount	13.03	81.62 × 19 ÷ 119, rounded once
BT-116 VAT category taxable amount	68.59	81.62 − 13.03, the remainder

BT-116V = BT-112 = BT-115 = 81.62. BT-131 Invoice line net amount is not used in the gross model: no conversion per line, no tolerance.

For information only: BT-146 Item net price = 1.799 ÷ 1.19 = 1.51176470… — the only value with endless decimals.

THE NET PRICE CONFIRMS THE VAT

1,000 × €24.95 incl. 7% VAT (#164)

	Net	VAT
4 dp ×	23,317.80	1,632.25
6 dp ✓	23,317.757	1,632.24
Extracted	23,317.76	1,632.24

BT-146 × BT-129 × 7%: with 4 dp the net price contradicts the extracted VAT, with 6 dp it confirms it.

Proposed check: Σ BT-146 × BT-129 ÷ BT-149 × rate gives BT-117 to the cent — possible only without a maximum

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AMENDMENT · PROPOSAL 2/2

Round only once (#163)

Only what is paid gets rounded, once: the amount due and the VAT.

Paid amounts and VAT — rounded once

- BT-115 Amount due for payment, in the Minor Unit (EUR: cent)
- BT-117 VAT category tax amount: once per VAT category
- BT-114 Rounding amount carries the rounding of BT-115
- Likewise BT-110, BT-111, BT-113, BT-172, BT-179, BT-183

Everything else: as calculated

- Prices, quantities, line amounts and sums: no maximum dp
- BT-116 VAT category taxable amount: exact, cents via CIUS
- Because the decimals needed grow with quantity and lines

WHAT CHANGES

- 6.5.14 applies to the whole chain, not just per rule
- Exact validation (decimal128); each rule keeps today's tolerance
- Error only outside it; a warning only where a rounding changes BT-117 or BT-112
- No requirement on software: neither a maximum nor a minimum of decimal places
- Syntax unchanged: amounts in UBL and CII are xsd:decimal

Status 16 Oct 2026: Update 4 in WG1 · VAT in cents, as Paul Simons asked; no warning for a rounding that does not change the result

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CALCULATION CHAIN · PROPOSED EN 16931-1:2026 AMENDMENT (#163)

From gross price via net price to VAT

No intermediate rounding — only BT-115 Amount due for payment and BT-117 VAT category tax amount are rounded, once.

INPUT & INTERMEDIATE VALUES

BT-146	**Item net price** = BT-148 − BT-147	free unrounded	± 0 as agreed
BT-131	**Invoice line net amount** = **BT-146** × BT-129 ÷ BT-149 + Σ BT-141 − Σ BT-136	free unrounded	± 0 exact
BT-106	**Sum of invoice line net amount** = Σ **BT-131**	free unrounded	± 0 exact sum

FINAL RESULTS

BT-116	**VAT category taxable amount** = Σ **BT-131** + Σ BT-99 − Σ BT-92	free unrounded	± 0 exact
BT-117	**VAT category tax amount** = **BT-116** × BT-119 ÷ 100	Minor Unit rounded once	± 0 once per VAT category
BT-110	**Invoice total VAT amount** = Σ **BT-117**	Minor Unit Σ of BT-117	± 0
BT-115	**Amount due for payment** = **BT-110** + **BT-109** Invoice total amount without VAT − **BT-113** Paid amount + **BT-114** Rounding amount	Minor Unit rounded once	± 0 BT-114 shows the rounding

ALLOWED DECIMAL PLACES

BT-148 Item gross price	free
BT-147 Item price discount	free
BT-146 Item net price	free
BT-129 Invoiced quantity	free
BT-149 Item price base quantity	free
BT-141 Invoice line charge or tax amount	free
BT-136 Invoice line allowance amount	free
BT-131 Invoice line net amount	free
BT-106 Sum of invoice line net amount	free
BT-99 Document level charge or tax amount	free
BT-92 Document level allowance amount	free
BT-119 VAT category rate (law)	2 dp

DECIMAL PLACES · FINAL VALUES

BT-116 VAT category taxable amount	free
BT-117 VAT category tax amount	Minor Unit
BT-110 Invoice total VAT amount	Minor Unit
BT-115 Amount due for payment	Minor Unit

Worst case with number of lines and quantity:
→ "How large does the deviation get?"

Red = rounded intermediate value Green = paid amount or VAT, rounded once Orange = rounded input value
dp = decimal places · Minor Unit = smallest currency unit, ISO 4217 (EUR: 2 dp) · ± for 100 lines of 100 units, 19% VAT
Rainbow = order of the calculation steps; a BT keeps its colour where it is reused further down

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WORST CASE · PROPOSED AMENDMENT (#163)

How large does the deviation get?

No factor: without intermediate rounding it stays at ± 0 — for any quantity and any number of lines.

Factor 1 · price rounding × quantity Q

BT-146 Item net price and **BT-131** Invoice line net amount stay exact — the quantity magnifies no error.

none

Factor 2 · number of lines n

BT-131 Invoice line net amount stays exact — it does not matter how many lines **BT-106** Sum of invoice line net amount and **BT-116** VAT category taxable amount add up.

none

NET DEVIATION **BT-106** Sum of invoice line net amount / **BT-116** VAT category taxable amount

Units per line (Q) → ↓ Lines (n)	100 units	10,000 units	1,000,000 units
1 line	Price × quantity ± 0 + rounding BT-131 ± 0 = ± 0	Price × quantity ± 0 + rounding BT-131 ± 0 = ± 0	Price × quantity ± 0 + rounding BT-131 ± 0 = ± 0
100 lines	± 0	± 0	± 0
10,000 lines	± 0	± 0	± 0

Assumption: all roundings go in the same direction, BT-149 Item price base quantity = 1, 19% VAT.

TOTAL · NET

± 0

Only BT-115 Amount due for payment and BT-117 VAT category tax amount are rounded, once.

100 lines of 10,000 units
Net ± 0 VAT ± 0

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IMPACT

What changes for existing implementations

Nothing that works today breaks — effort only where accuracy was impossible.

Who	What changes	Effort
Issuers: net prices, 2 dp	nothing	none
Receivers with 2 dp	round precise values once on import	small, only for precise invoices
Gross-price invoices	new “plus VAT” terms (#164)	new core feature; no correct way existed
Validators	exact decimal arithmetic (decimal128)	test documents exist (Svanton)
Syntaxes UBL and CII	none: amounts are xsd:decimal	none

Removing a maximum is not a minimum: every invoice that is conformant today stays conformant, at most with a warning (#163).

Gross-price invoices never worked in the core: BMF letter para. 18; workaround only until 31 Dec 2027 (para. 19).

WHAT STAYS

- Invoices with 2 decimal places stay conformant
- The net model is unchanged
- BT-110, BT-115, BT-117 booked in cents as today
- Today's tolerances remain; warning only if the result changes

Impact study (Cyrille Sautereau): does a system round a precise value on import — or reject the invoice?

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GROSS MODEL (#164) · WHERE

Why the gross model belongs in the core

Shops and cash registers have no prior agreement with their buyers.

Only the core reaches every receiver

- No acceptance needed for EN 16931 invoices (new Art. 232, ViDA)
- DRR accepts EN-compliant data (new Art. 263(3), ViDA)
- A CIUS such as XRechnung can only restrict (7.1, Table 29)

The alternatives do not close the gap

- Advanced profile or extension: only with prior agreement
- Simplified invoice: small amounts only (Art. 220a; DE ≤ €250)
- One line per rate + attachment: until 31 Dec 2027

BMF LETTER · 15 OCT 2026

- Para. 17: VAT to the cent; net amount = consideration minus VAT
- Para. 18: core model and tax rounding rule cannot both be met
- Para. 18: a format based on gross prices is recommended
- Para. 19: XRechnung and ZUGFeRD have no gross format yet

Use cases: fixed book prices · receipt → invoice · retail and marketplaces · fuel · statutory gross prices

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STATUS · 20 OCT 2026

Where the discussion stands

Much is agreed already — four points remain open.

Where the proposals meet

- VAT in the Minor Unit, rounded once per VAT category
- "Plus VAT" terms and an indicator in the core
- Precise line values and more price decimals than today
- Proposed by Paul Simons and Cyrille Sautereau, 16–19 Oct

Still open

- One line amount, or a second precise one (BT-131NR)
- A maximum of 8 decimals, or none
- BT-116 VAT category taxable amount: exact, or in cents
- VAT first: a rule in the core, or guidance and CIUS

IN THE END: ACCURATE DATA FOR ALL

- Conformant to the EN and compliant with national VAT law
- A tolerance for those who accept it, no barrier for others
- Receivers round on import, not reject (new Art. 232, ViDA)
- Software states its precision (decimal32, 64, 128); the market chooses

Conformant ≠ compliant: for the same supply, the example of Paul Simons gives €1.20 or €1.21; the EN tolerates it, German law has no tolerance

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DECISIONS

Two decisions for WG1

Both keep every working implementation conformant.

1 · Gross model in the core (#164)

- “Plus VAT” terms such as BT-146V, BT-131V, BT-116V
- VAT first: BT-117 rounded once, BT-116 = BT-116V − BT-117
- No net intermediate values; BT-146 for information

2 · No forced rounding (#163)

- No maximum dp for prices, quantities, intermediate values
- Only paid amounts and VAT rounded, once; BT-114 shows it
- Today's tolerances stay; warning only if the result changes

NEXT STEPS

- Confirm VAT first as the European method with TAXUD
- Ask TAXUD: unit price exclusive of VAT optional for gross invoices?
- Test documents for the decimal levels decimal32, 64 and 128
- Impact test with ERP and e-invoicing vendors

Issues: github.com/ConnectingEurope/CEN-TC434-WG1/issues/164 · /163

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CONTACT

Talk to me

Slides as PPTX

Svante Schubert
Syntax bindings · validation · tooling
Editor of the syntax bindings, CEN/TC 434 WG3

Follow & join in

- WG1 issue #164 · gross price model
- WG1 issue #163 · round only once
- BMF letter of 15 Oct 2026, section 2.5
- Svanton: exact decimal arithmetic for Saxon

Thank you for your attention — I look forward to the discussion in WG1.

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